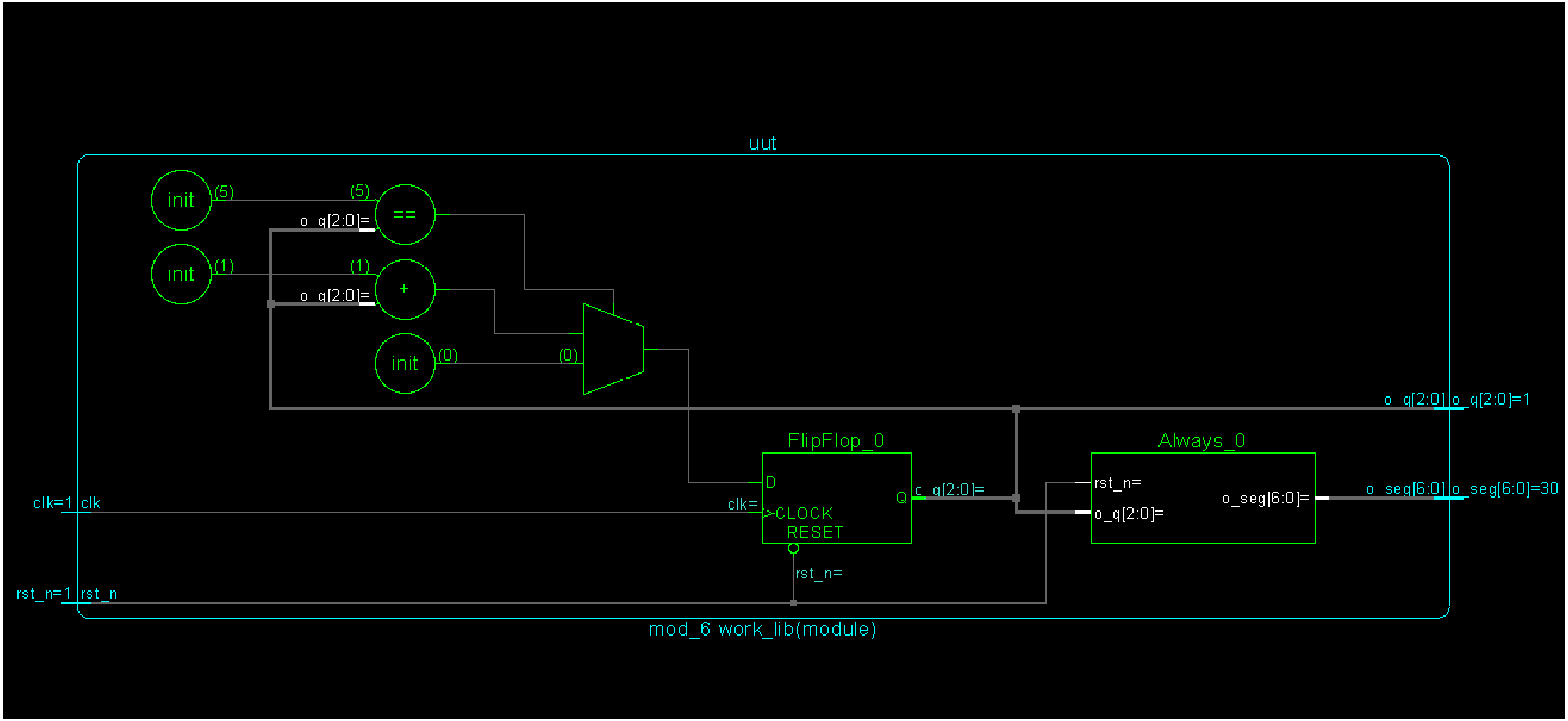
SoC 102: Design

중간점검: Basic Sequential Logic Circuit / Synthesis / PNR

MOD-6

성남폴리텍 하이테크 과정_김동섭

Design: Schematic



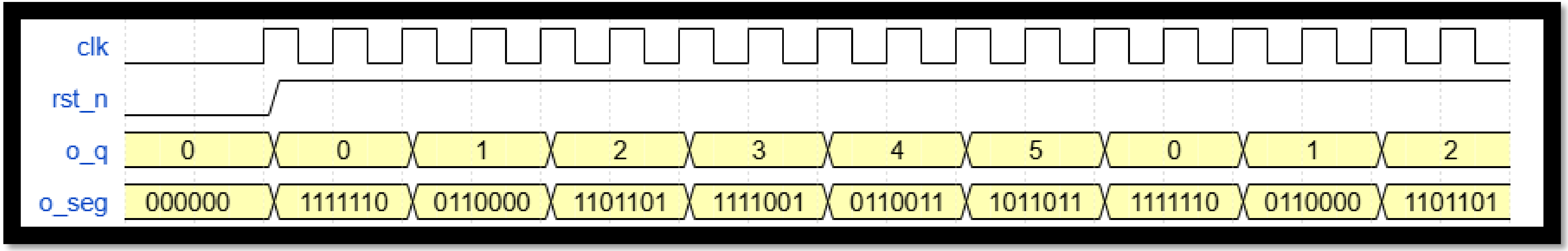
Design: RTL Coding

```
`timescale 1ns / 1ps

module mod_6 (
        // port list
        clk                             ,
        rst_n                           ,
        o_q                             ,
        o_seg
);

        // port declaration
        input                   clk     ;
        input                   rst_n   ;
        output  [2:0]           o_q     ;
        output  [6:0]           o_seg   ;

        // modeling
        reg     [2:0]           o_q     ;
        reg     [6:0]           o_seg   ;
        always @(posedge clk or negedge rst_n) begin
                if (!rst_n) begin
                    o_q <= 3'b000;
                end else begin

                        if (o_q == 3'b101) begin
                            o_q <= 3'b000;
                        end else begin
                            o_q <= o_q + 1'b1;
                        end
                end
        end

        // 7_seg Decoder Logic
        always @(*) begin
            if (!rst_n) begin
                o_seg = 7'b0000000;
            end else begin
                case (o_q)
                        3'b000  : o_seg = 7'b1111110;   //0
                        3'b001  : o_seg = 7'b0110000;   //1
                        3'b010  : o_seg = 7'b1101101;   //2
                        3'b011  : o_seg = 7'b1111001;   //3
                        3'b100  : o_seg = 7'b0110011;   //4
                        3'b101  : o_seg = 7'b1011011;   //5
                        default : o_seg = 7'bz;         //disabled
                endcase
            end
        end


        endmodule
```

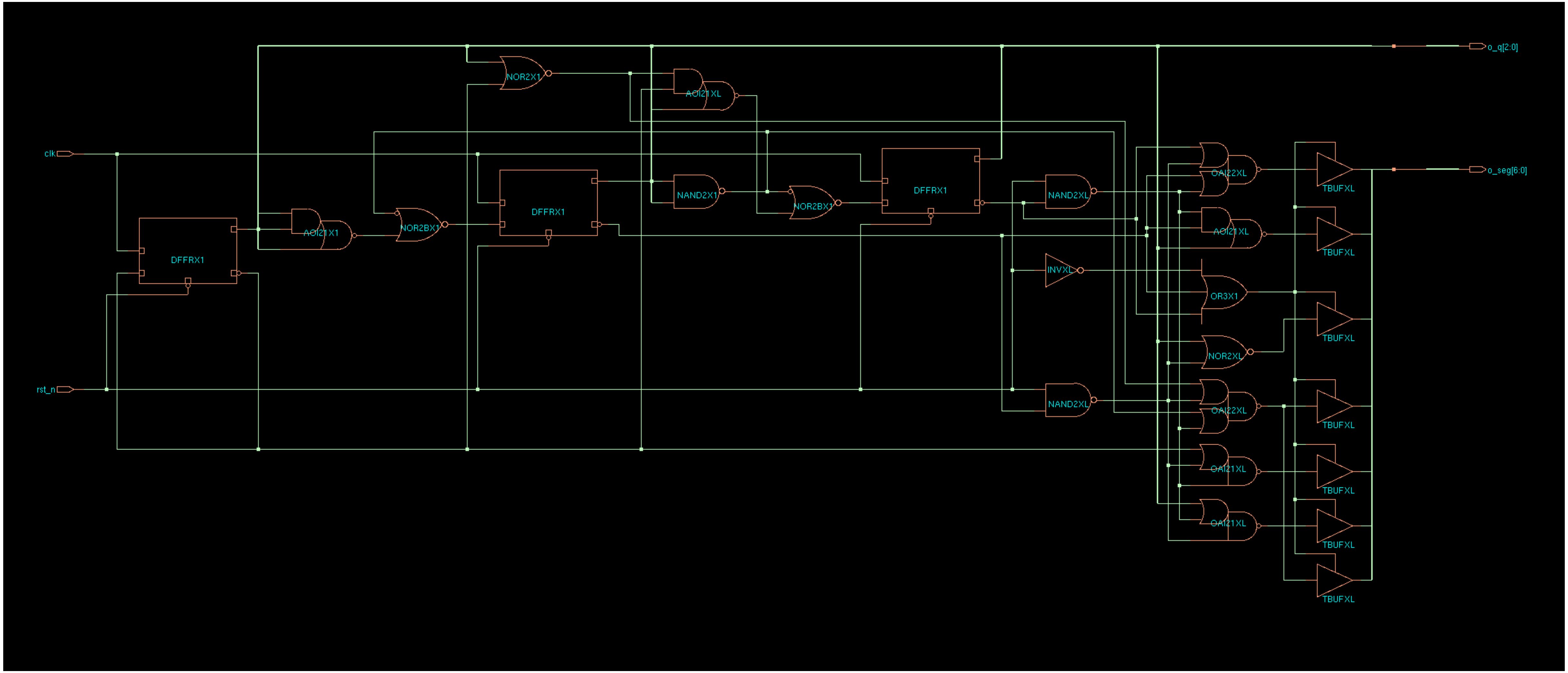
Testbench

```
`timescale 1ns / 1ps

        module tb_mod_6();

        // port list
        reg             clk             ;
        reg             rst_n           ;
        wire    [2:0]   o_q             ;
        wire    [6:0]   o_seg           ;

        // DUT instantiation
        mod_6 uut (
                .clk            (clk            ),
                .rst_n          (rst_n          ),
                .o_q            (o_q            ),
                .o_seg          (o_seg          )
        );

        // clock gen
        always #5 clk = ~clk;


        initial begin

        #10;
        clk     = 0;
        rst_n   = 0;


        #10;
        rst_n   = 1;


        #200;


        $finish;

        end

        endmodule
```

Design: Waveform



Synthesis



clk
rst_n
DFFRX1
AOI21X1
NOR2BX1
NOR2X1
DFFRX1
AOI21XL
NAND2X1
NOR2BX1
DFFRX1
NAND2XL
INVXL
NAND2XL
OAI22XL
AOI21XL
OR3X1
NOR2XL
OAI22XL
OAI21XL
OAI21XL
TBUFXL
o_q[2:0]
o_seg[6:0]

Synthesis

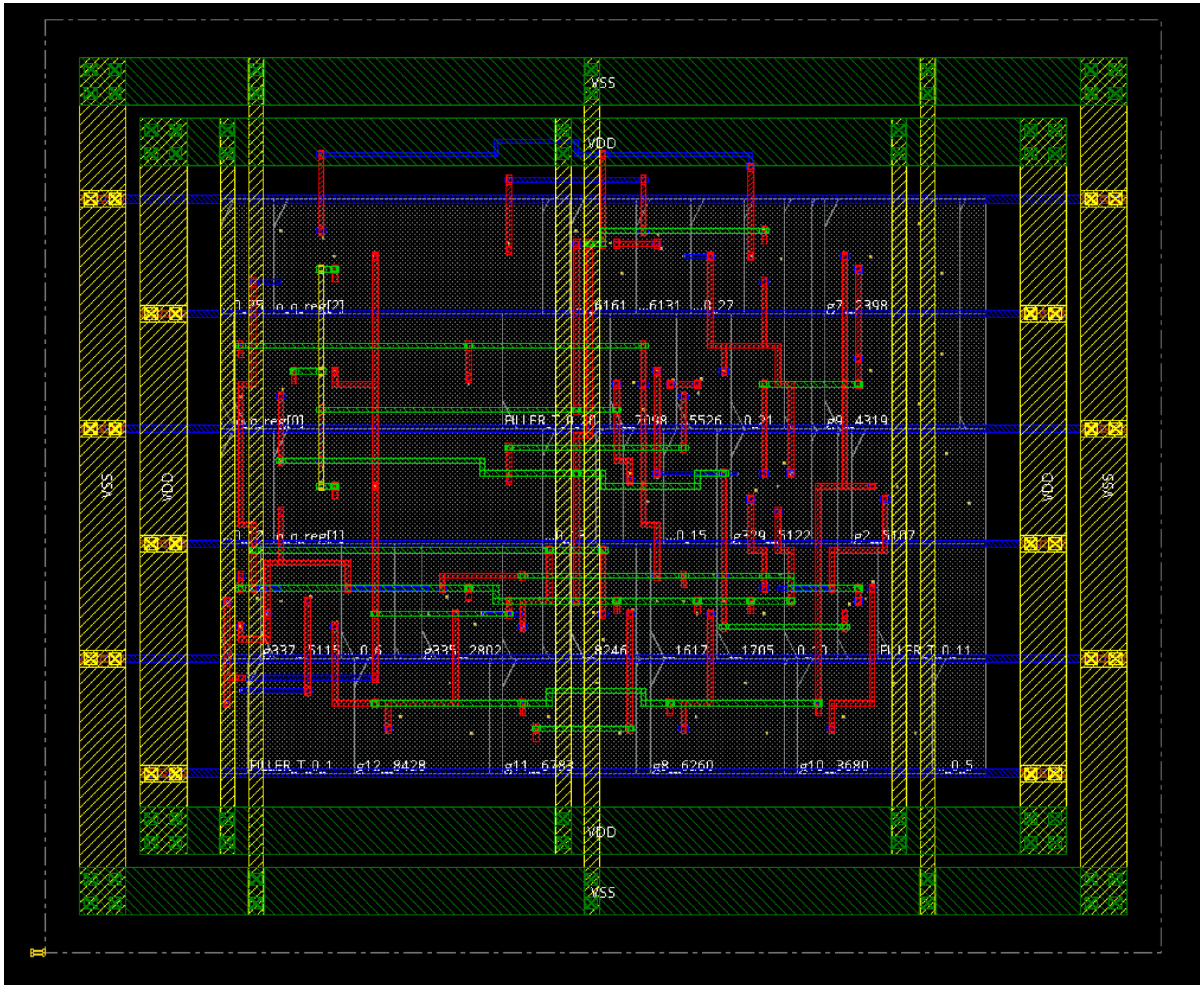
```
module mod_6(clk, rst_n, o_q, o_seg);
  input clk, rst_n;
  output [2:0] o_q;
  output [6:0] o_seg;
  wire clk, rst_n;
  wire [2:0] o_q;
  wire [6:0] o_seg;
  wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
  wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
  wire n_18, n_20, n_23;
  TBUFXL g7__2398(.A (n_20), .OE (n_23), .Y (o_seg[6]));
  TBUFXL g2__5107(.A (n_11), .OE (n_23), .Y (o_seg[5]));
  TBUFXL g8__6260(.A (n_15), .OE (n_23), .Y (o_seg[4]));
  TBUFXL g9__4319(.A (n_20), .OE (n_23), .Y (o_seg[3]));
  TBUFXL g12__8428(.A (n_14), .OE (n_23), .Y (o_seg[0]));
  NOR2BX1 g331__5526(.AN (n_10), .B (n_7), .Y (n_18));
  TBUFXL g11__6783(.A (n_9), .OE (n_23), .Y (o_seg[1]));
  TBUFXL g10__3680(.A (n_3), .OE (n_23), .Y (o_seg[2]));
  OAI21XL g334__1617(.A0 (n_6), .A1 (n_13), .B0 (n_12), .Y (n_15));
  OAI22XL g335__2802(.A0 (n_2), .A1 (n_13), .B0 (n_8), .B1 (n_12), .Y
       (n_14));
  OAI21XL g333__1705(.A0 (o_q[0]), .A1 (n_12), .B0 (n_13), .Y (n_11));
  OAI22XL g329__5122(.A0 (n_5), .A1 (n_13), .B0 (n_10), .B1 (n_12), .Y
       (n_20));
  AOI21XL g339__8246(.A0 (o_q[0]), .A1 (n_8), .B0 (n_12), .Y (n_9));
  AOI21XL g340__7098(.A0 (o_q[1]), .A1 (n_6), .B0 (n_5), .Y (n_7));
  NOR2BX1 g332__6131(.AN (n_10), .B (n_1), .Y (n_4));
  NOR2XL g336__1881(.A (o_q[0]), .B (n_13), .Y (n_3));
  OR3X1 g337__5115(.A (n_0), .B (n_8), .C (n_2), .Y (n_23));
  NOR2X1 g343__7482(.A (o_q[1]), .B (n_6), .Y (n_5));
  NAND2XL g345__4733(.A (rst_n), .B (n_8), .Y (n_13));
  AOI21X1 g338__6161(.A0 (o_q[1]), .A1 (o_q[0]), .B0 (o_q[2]), .Y
       (n_1));
  NAND2XL g344__9315(.A (rst_n), .B (n_2), .Y (n_12));
  NAND2X1 g341__9945(.A (o_q[2]), .B (o_q[0]), .Y (n_10));
  INVXL g347(.A (rst_n), .Y (n_0));
  DFFRX1 \o_q_reg[2] (.RN (rst_n), .CK (clk), .D (n_4), .Q (o_q[2]),
       .QN (n_8));
  DFFRX1 \o_q_reg[1] (.RN (rst_n), .CK (clk), .D (n_18), .Q (o_q[1]),
       .QN (n_2));
  DFFRX1 \o_q_reg[0] (.RN (rst_n), .CK (clk), .D (n_6), .Q (o_q[0]),
       .QN (n_6));
endmodule
```

Mod_6_netlist.v – Gate Level Netlist

```
set sdc_version 2.0

set_units -capacitance 1000fF
set_units -time 1000ps

# Set the current design
current_design mod_6

create_clock -name "clk" -period 10.0 -waveform {0.0 5.0} [get_ports clk]
set_clock_transition 0.1 [get_clocks clk]
set_clock_gating_check -setup 0.0
set_wire_load_mode "enclosed"
set_clock_uncertainty -setup 0.01 [get_ports clk]
set_clock_uncertainty -hold 0.01 [get_ports clk]
```

Mod_6_final.sdc – Synopsys Design Constraints

APR




```
VERIFY DRC ...... Starting Verification
VERIFY DRC ...... Initializing
VERIFY DRC ...... Deleting Existing Violations
VERIFY DRC ...... Creating Sub-Areas
VERIFY DRC ...... Using new threading
VERIFY DRC ...... Sub-Area: {0.000 0.000 16.600 13.870} 1 of 1
VERIFY DRC ...... Sub-Area : 1 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU TIME: 0:00:00.0  ELAPSED TIME: 0:00:00.0  MEM: 264.1M) ***

@innovus 31> set_db check_drc_area {0 0 0 0}
@innovus 32> gui_select -point {3.80200 -0.48200}
@innovus 33>
```

```
******** Start: VERIFY CONNECTIVITY ********
Start Time: Thu Jul  9 18:38:19 2026

Design Name: mod_6
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (16.6000, 13.8700)
Error Limit = 1000; Warning Limit = 50
Check all nets
**WARN: (IMPVFC-97):    IO pin clk of net clk has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin rst_n of net rst_n has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_q[2] of net o_q[2] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_q[1] of net o_q[1] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_q[0] of net o_q[0] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[6] of net o_seg[6] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[5] of net o_seg[5] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[4] of net o_seg[4] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[3] of net o_seg[3] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[2] of net o_seg[2] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[1] of net o_seg[1] has not been assigned. Please make sure it is assigned and rerun check_connectivity.
**WARN: (IMPVFC-97):    IO pin o_seg[0] of net o_seg[0] has not been assigned. Please make sure it is assigned and rerun check_connectivity.

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Thu Jul  9 18:38:19 2026
Time Elapsed: 0:00:00.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.0  MEM: 0.000M)
```

자기평가

	수강여부	숙련도(상/중/하)
회로이론		
전자회로	O	하
디지털논리회로	O	하
디지털 시스템설계		
신호와 시스템		
HDL 설계언어(Verilog/VHDL)		
컴퓨터구조론		
디지털통신공학		
VLSI 설계		
C언어	O	하
마이크로프로세서프로그래밍		

- 본전공 → 반도체설계과
- 직무희망 / 관심영역 → ANALOG LAYOUT / PD
- 방학 학습계획 → (유튜브) 뽕교수 강의
→ Digital Design
→ Digital Systems Design Using Verilog
- Git 주소 → https://github.com/ehdtjq1289